

ARTICLE

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Multiscale spatial dependency of small mammal diversity patterns and their determinants

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Abstract

Animals perceive their ecosystems at multiple spatial scales such that the pattern–process relationships that determine their ecological and evolutionary opportunities are also spatially variable. Understanding how spatial scale impacts our observations of ecological and evolutionary patterns and processes is therefore crucial for effective biodiversity management. In this study, using a comprehensive dataset of small mammals from primary field surveys supplemented with literature data across Kenya, we investigated the effects of spatial scale on the distribution patterns of taxonomic, phylogenetic, and functional diversity indices. We also tested how relationships between taxonomic, phylogenetic, and functional diversity and various environmental and anthropogenic variables change across spatial scales. The results showed that with increasing spatial scale, species richness and phylogenetic and functional richness consistently increased. Community-wide mean pairwise distance indices for phylogenetic and functional dimensions also increased, while fine-scale mean nearest taxon distance indices decreased. Notably, community structure indices exhibited nonmonotonic patterns across spatial scales. The associations between diversity patterns and environmental and human covariates increased with scale but were more variable and complex across regions. These findings highlight a negative correlation between spatial scale and phylogenetic and

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functional divergence while reinforcing the species-area hypotheses for richness-dependent diversity indices. Identifying the optimal scales of field studies and conservational actions must factor in the species group of interest and the geographical context.

KEYWORDS

biogeography, community structure, diversity and distribution, functional diversity, phylogenetic diversity, small mammals, spatial scale, species richness

INTRODUCTION

Variable species' perceptions of niches, habitats, and ecosystems across spatial scales regulate their ecological–evolutionary opportunities and diversity pattern–process relationships (Chase et al., 2019; Holland et al., 2004; Levin, 1992; Rahbek, 2005; Wiens, 1989). However, despite varying species diversity and distribution patterns across spatial scales being of core interest in the fields of ecology, evolutionary biology, and biodiversity (Blicharska et al., 2019; Díaz et al., 2005; McGill, 2010), unifying the effects of spatial scale on species ecological functions and evolutionary histories remains elusive (Estes et al., 2018) because of geographical unevenness, and increasingly intractable human-driven ecological changes. Extending multiscale investigations of species diversity patterns and their determinants to underexplored regions, such as the Afrotropic realm, would enhance our understanding of scale influences on ecological and evolutionary processes, while improving biodiversity monitoring, assessment, and conservation efforts.

Despite significant progress in species-level ecological and evolutionary research explaining the high mammalian diversity of the Afrotropics, these advances remain misaligned with the predominantly national-scale biodiversity conservation programs in the region (Wilson et al., 2007). Refined insights at the national level are urgently needed, particularly regarding how field sampling scale affects observed diversity and distribution patterns. For instance, while water and energy availability dynamics are recognized as primary drivers of mammalian diversity and distribution in the tropics (Brown, 2014), the extent to which these influences are spatially constrained—especially in the context of evolutionary enclaves and geographic regionalization (Brown, 2014; Hawkins et al., 2003; Menge & Sutherland, 1976)—remains poorly understood. The interpretation of biotic and abiotic factors driving community structure is inherently spatially constrained. For instance, the abiotic filtering principle emphasizes the role of environmental effects at larger scales where species must adapt to similar conditions through the convergence of ecological functions, leading

to communities dominated by more closely related species than would be expected by chance (Hardin, 1960; HilleRisLambers et al., 2012). Biotic filtering, on the other hand, emphasizes the effect of inter-taxa competition at local scales, systematically filtering out species with similar ecosystem functions, leading to communities composed of less closely related species than would be expected by chance (de la Sancha et al., 2023; Rahbek, 2005; Swenson et al., 2007; Tilman, 2004).

Investigating how the extent of the study area, the choice of sampling scale, and the evolutionary depth of applied diversity metrics influence observed patterns and processes can provide a holistic understanding of the relative impacts of abiotic and biotic factors on ecological dynamics and biodiversity vulnerabilities to long-term habitat and ecosystem changes (de la Sancha et al., 2017; Hortal et al., 2010; Leao-Pires et al., 2018; Magnusson et al., 2020; Whittaker et al., 2001; Wiegand et al., 2017). The scale at which various environmental variables (climate, habitat productivity, topographic heterogeneity) most strongly affect diversity–predictor relationships is closely related to the variables' heterogeneity (Macarthur & Macarthur, 1961). For example, while the influences of climate and vegetation on vertebrate community structure are mostly stronger at larger regional and landscape scales where they are more heterogeneous, their effects are weaker or may even disappear at local scales due to low heterogeneity (Field et al., 2008; Kent et al., 2011). Moreover, because recent species divergences and evolutionary processes tend to be relatively localized, they are more likely to result in phylogenetic clustering at smaller spatial scales. In contrast, deeper divergences are more likely to result in phylogenetic overdispersion at larger scales, where related lineages have had sufficient time to disperse and colonize broader areas (Swenson et al., 2007). The consistency of these associations across spatial scales remains untested extensively across broader spatial scale gradients. Moreover, because most mammalian fauna underwent rapid evolutionary radiation in the tropics, phylogenetic and functional diversification and competitive dynamics may have driven phylogenetic overdispersion or clustering at large spatial scales (Cardillo, 2011). However,

there are hardly finer scale investigations at the habitat and microhabitat levels to critically test the limits of such effects and extend the relevancies of practical application.

Defining the optimal spatial scale at which to conduct field surveys is critical in biodiversity assessment and prioritization to match factors such as fund availability and the extent and severity of human encroachment into wildlife areas (Chase et al., 2019; Fahrig et al., 2022; Gonzalez et al., 2016; Henle et al., 2014). For example, species diversity can increase at local scales in response to conservation efforts such as afforestation or edge effects but still be observed to decline at regional or global scales (Chaves et al., 2022; de la Sancha et al., 2014, 2020; Gonzalez et al., 2016; Sax & Gaines, 2003; Thomas, 2013; Vellend et al., 2013). Disentangling how human-caused species losses that occur at local scales build up to the global scale, or vice versa, is vital in navigating complex conservation scenarios where the targeting of keystone species or vital ecosystems may focus on particular patches or landscape fragments (de la Sancha et al., 2021; de la Sancha & Boyle, 2019; Fischer & Lindenmayer, 2007; Onditi, Li, et al., 2021), while programs emphasizing species or community richness are more likely to concentrate on conserving more expansive landscape mosaics (Chase et al., 2019; Wiens, 1989). The spatial scale of biodiversity measurement also determines how areas are prioritized as hotspots of diversity, without which conservation needs may be ecologically desynchronized with nationally designated conservation areas (Díaz et al., 2006; Flynn et al., 2009; Graham et al., 2018; Mace et al., 2003).

We analyzed a comprehensive dataset of small mammals from Kenya, assessing the variation of multiple diversity metrics across several sampling scales: trapline, transect, locality, and site. Based on the practical necessity to progress tailored ecological studies to national contexts highlighted in the preceding paragraphs, Kenya provides an ideal geographic focus for serving as the eastern range limit for many species with core distributions in the Central African Congo–Guinean forests and the Eastern Afrotropical Biodiversity hotspot. Our study addressed two main questions: (1) How do spatial scale variations influence observed small mammal diversity and distribution patterns? (2) How do the associations between small mammal diversity and environmental factors differ at different spatial scales? To align our field observations with broader biodiversity patterns for management relevance, we supplemented our analysis with a nationwide dataset of species composition, phylogenetics, and functional traits following literature review. We hypothesized a consistent increase in species richness (SR) and related diversity indices at larger spatial scales (Chase et al., 2019; Henle et al., 2014; Rahbek, 2005), phylogenetically overdispersed assemblages at local scales (species less closely related

than would be expected by chance), phylogenetically clustered assemblages at broader scales (Gotelli et al., 2010; McGill, 2010), and stronger correlations between environmental variables and diversity patterns at larger scales (Cardillo et al., 2008; Chase et al., 2018; Hortal et al., 2010; Ricklefs, 2004). These hypotheses are primarily based on the ecological rationale that functionally dissimilar species tend to aggregate at smaller spatial scales due to reduced direct competition. In contrast, phylogenetically related species, which often share ecological niches due to their common evolutionary history, may experience more intense competition at such scales. As a result, they either avoid each other at small scales or co-occur at larger scales, where significant habitat heterogeneity facilitates their coexistence (Schoener, 2009).

METHODS

Study area and sampling design

The study is based on fieldwork conducted in Kenya between September 6, 2015, and November 14, 2018. Over this period, four ca. 2-month-long surveys were conducted across 10 sites, which were selected following reconnaissance surveys in preceding years. In this study, we clustered these sites into six study sites based on geographical proximity: Mount Kenya, Chyulu Hills and Tsavo East and West National Parks, Shimba Hills National Park and Gongoni Forest Reserve, Loita Hills, Mau Forest, and Kakamega Forest and Yala Swamp (Figure 1).

The sites are characterized by prominent bioclimatic gradients in climate and geomorphology and span key ecosystems in Kenya (coastal forests, eastern Afrotropical forests, semiarid savanna woodlands-grasslands, and Victoria Basin forest-savanna), thus providing an ideal ecological system for obtaining comparative insights into small mammal community ecology dynamics. We partitioned the sampled units into 19 localities, 65 transects, and 255 traplines (Onditi et al., 2025: data tab. 1). The sampling scales were modified using the strictly nested quadrat (SNQ) method (Leitner & Rosenzweig, 1997; Storch & Chiarucci, 2016) to treat sites, localities, transects, and traplines as having varying cell sizes/grains. As such, traplines ($\bar{x}$: 1.9 km²) were nested within transects ($\bar{x}$: 6.3 km²), which were in turn nested within localities ($\bar{x}$: 413 km²), which were nested within sites ($\bar{x}$: 5057 km²).

For animal trapping, we used a combination of sherman traps, snap traps, and pitfall traps primarily targeting animals in the Rodentia and Eulipotyphla orders. Traps were set to optimize diverse habitats (including primary and secondary forests, farmlands, grasslands, and ecotones) to optimize trapping success (see Musila, Chen,

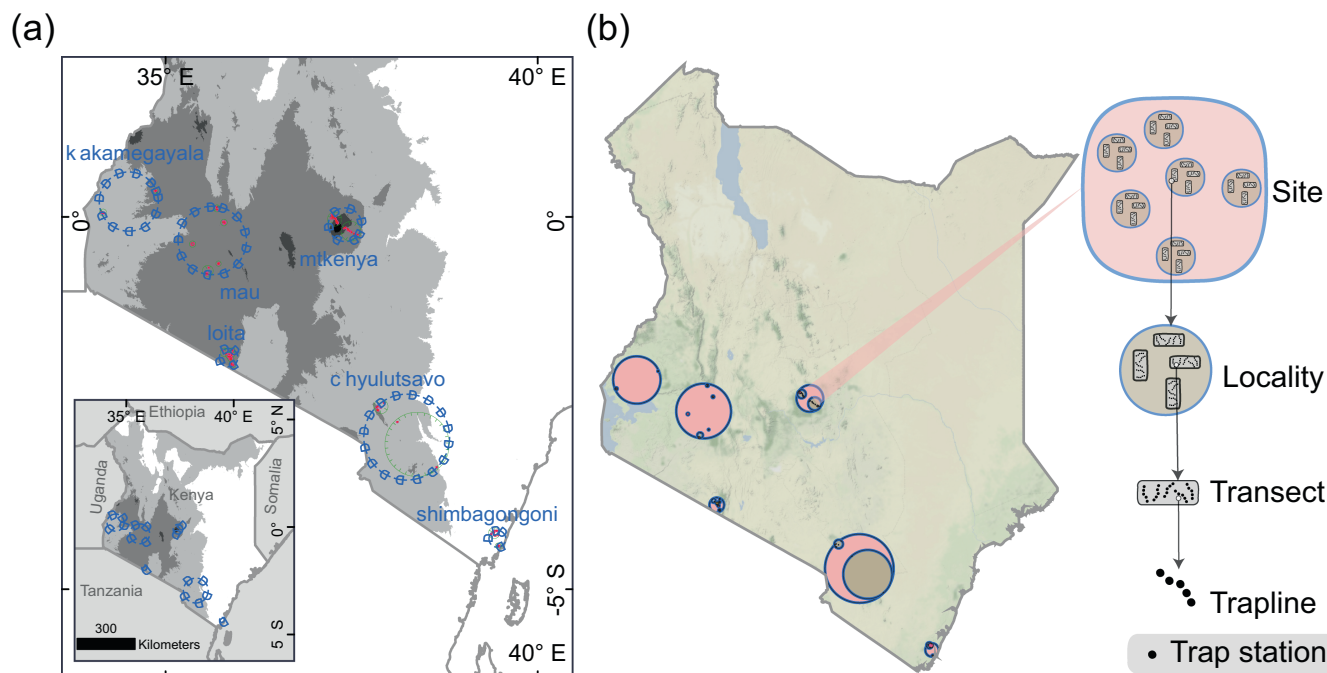


FIGURE 1 Topographical map of Kenya showing the primary field surveys, sampling sites, and sampling scheme. (a) The locations of the primary sampling sites, with the inset indicating the localities within the administrative boundary of Kenya. (b) The sampling scheme employed in the study, with the nested nature of spatial scales illustrated.

et al., 2019; Onditi et al., 2020, 2022 for more exhaustive trapping descriptions). The animal handling procedures adhered to Kenya's wildlife research regulations and the established guidelines for using wild animals in research (Sikes & The Animal Care Use Committee of the American Society of Mammalogists, 2016; Underwood & Anthony, 2020). The research was approved by the Kenya Wildlife Service Research and Ethics Committee (permit number: KWS/BRM/5001).

DNA extraction and sequencing, phylogenetic analysis, and species delimitation

We first assigned taxonomic identifications based on external morphological examination to at least generic levels following various taxonomic works (Carleton & Musser, 2013; Happold & Happold, 2013; Hutterer, 2005; Monadjem et al., 2015; Musser & Carleton, 2005) and following previous works on Kenya's small mammals during field surveys (Musila, Chen, et al., 2019; Musila, Monadjem, et al., 2019; Onditi et al., 2020; Onditi, Demos, et al., 2021). However, because field identification can be problematic, we integrated morphological and distribution characterization and DNA barcoding to identify molecular operational taxonomic units (MOTUs) corresponding to available species names and potentially

undescribed species. The MOTUs capture genetic variation within communities, and although they may not always align with species classifications, they reflect the underlying genetic diversity that can impact ecological interactions and evolutionary processes.

We sequenced the mitochondrial *cytochrome b* gene (1140 base pairs long) from 3255 samples. To this end, total DNA was extracted from muscle or liver tissue preserved at -80°C in absolute (99.6%) ethanol following the sodium dodecyl sulfate procedure (Sambrook et al., 1989). The PCR template included 1.5 μL of total DNA template, 0.5 μL of gene-specific primers, 10 μL of 2 $\times$ Power TaqPCR Master Mix, and 8.5 μL of ultrapure water, resulting in a total volume of 20 μL of reaction reagents per sample. The mitochondrial *cytochrome b* gene was amplified using the primers L14724 (5'-GGACTTATGACATGAAAAATCATCGTTG-3') and H14139 (5'-GATCCCCATTCTGTTTACAAGAC-3') (Kocher et al., 1989). The PCR included 40 cycles of 30 s of denaturation at 95.0°C , 30 s of annealing at 59.0°C , and 60 s of extension at 72.0°C . The amplified product was sequenced in both directions using the same primers on an ABI sequencer (Applied Biosystems: www.thermofisher.com). The resulting chromatograms were de novo assembled using Geneious Prime 2022.0.1 (www.geneious.com) with default settings and aligned with MAFFT version 7 (Katoh & Standley, 2013). The alignment was analyzed in DnaSP version 6 (Rozas

et al., 2017) to retrieve haplotypes that were then used to construct phylogenetic trees for species delimitation.

For phylogenetic analyses, we used the maximum likelihood (ML) and Bayesian inference (BI) methods based on the GTR + I + G substitution model, which was selected in ModelFinder (Kalyaanamoorthy et al., 2017) as the best fit under a BI criterion. The ML analysis was performed in IQ-TREE version 2.3.2 (Nguyen et al., 2015), with branch support estimated from 100,000 ultrafast bootstrap replicates (Minh et al., 2013). The BI analysis was performed in BEAST (Bouckaert et al., 2019) with two independent runs, each constituting 100 million Markov chain Monte Carlo (MCMC) chains sampled at 10,000 intervals. The results were visualized in Tracer to determine sampling adequacy (effective sample size values >200), and the log and tree files from the two runs were combined using LogCombiner (Rambaut & Drummond, 2024) and annotated using the maximum clade credibility method in TreeAnnotator (Rambaut & Drummond, 2024) with a 10% burn-in.

The resulting ML and BI trees and haplotype-derived alignment were used for species delimitation using tree-based and distance-based methods that could reliably identify common species units without prior sample assignment to taxonomic units. We used the branch-cutting method (BCUT; Ondřej, 2018), the multirate Poisson tree processes algorithm (mPTP; Kapli et al., 2017), and the single-threshold general mixed Yule coalescent model (GYMC; Fujisawa & Barraclough, 2013) for tree-based delimitation. The ML tree was used for BCUT and mPTP, and the BI tree was used for GMYC. The BCUT analysis was performed in R version 4.1.2 (R Core Team, 2024) with functions provided by the author (Ondřej, 2018), GMYC in R using "splits" version 1.0.19 (Ezard et al., 2017), and mPTP using command-line options. The mPTP implementation involved four MCMC runs of 500 million generations, each sampled every 50,000 runs with a 10% burn-in, with convergence confirmed from the combined likelihood plot. The distance-based delimitation was performed using assemble species by automatic partitioning (ASAP) (Puillandre et al., 2021: <https://bioinfo.mnhn.fr/abi/public/asap/asapweb.html>) with the haplotype alignment as input, the K80 Kimura distance measure, and all other settings at default. As each delimitation method proposed different clade clustering schemes—ASAP 84 MOTUs, BCUT 85, GMYC 134, and mPTP 107—we resolved a consensus clustering scheme comprising 139 MOTUs factoring the delimitation results, external morphological and craniodental characters, and BLAST (Altschul et al., 1990) search matches, with MOTU names extracted from the latest mammal checklists (Hutterer, 2005; Musila, Monadjem, et al., 2019; Musser & Carleton, 2005; Wilson, Lacher, et al., 2019; Wilson, Mittermeier, et al., 2019).

We reconstructed a time-calibrated phylogeny in BEAST using the MOTU alignment for estimating phylogenetic indices. Divergence time calibrations were based on the most recent common ancestors of select clades as specified in Appendix S1: Table S1. Other parameters were the same as those used in the BI tree reconstruction. The resulting phylogenetic tree revealed well-resolved internal (order, family, and genus) and terminal relationships (Appendix S1: Figure S1).

Assembly of species functional traits

We estimated functional diversity indices using external morphological characteristics recorded in the field, complemented with diet and activity traits extracted from the literature (Faurby et al., 2018; Jones et al., 2009; Wilman et al., 2014). We selected traits spanning anatomical, life history, diet, and activity characteristics based on their intrinsic utility in mammalian ecological and evolutionary opportunities—resource (e.g., place, food, and time) acquisition and usage strategies (Jones et al., 2009; Pianka, 2011; Wilman et al., 2014). The external body measurements included body mass in grams, head-body length, tail length, hindfoot length, and ear length in centimeters. The diet traits included proportions of the species diet comprised of invertebrates, mammals and birds, herptiles, fish, scavenging, fruit, nectar, seeds, and plants. The activity traits included binary scores for whether a species was nocturnal, crepuscular, and diurnal. To assess trait redundancy and handle the unequal contribution of individual traits to the community trait space, we used principal components analysis (PCA) to transform the field-recorded traits into orthogonal axes and retained the first three principal components (PCs), which together explained 98.8% of the variance as the new external morphology traits. We also applied PCA to the 10 diet traits and retained the first five PCs, which explained 99.2% of the variance. The final functional traits included three PCs representing external morphological features, five PCs representing diet, and three activity pattern categories (Onditi et al., 2025: data tab. 2). The approach of using PCA PC scores as new trait representatives offers a clear framework for comparing the ecosystem functions of species within communities, as the orthogonal axes in trait space provide a more straightforward comparison than individual traits alone. To ensure comparable contributions of individual variables and prevent related traits from dominating the multi-trait dissimilarity analysis, we used the "gawdis" R package version 0.1.5 (de Bello et al., 2020) to group associated variables.

Diversity index estimation

We used multiple diversity indices spanning the richness, divergence, and community structure of taxonomic, functional, and phylogenetic dimensions to investigate how species diversity within communities responds to variations in the spatial scale of sampling more comprehensively. Taxonomic diversity measures the number and distribution of species, phylogenetic diversity highlights evolutionary relationships, and functional diversity assesses species' ecological roles based on biological traits. The richness, divergence, and community structure aspects capture dynamics of species–species and species–environment interactions that are key to ecologically stable ecosystems. We estimated diversity indices for every field sampling scale, including the traplines, transects, localities, and sites, and explored inter-diversity correlations using PCA and Pearson correlation coefficient (r) to estimate whether and how they varied across spatial scales (Appendix S1: Figures S4–S6).

Richness indices

As a baseline index, we calculated SR from the sum of unique “species” at the sampled sites using the “specnumber” function in “vegan” version 2.6-4 (Oksanen et al., 2019). For phylogenetic richness, we used the phylogenetic diversity index from Faith (1992) and the phylogenetic measures of SR of Helmus et al. (2007). Both indices represent phylogenetic richness (PD^{RIC}) and measure the total evolutionary history based on the sum of branch lengths connecting all species, providing vital information regarding biodiversity management, such as a community's resilience to disturbance (Chao et al., 2016; Tucker et al., 2017). For functional richness (FD^{RIC}), we used the incidence-distance-based functional richness index from Vileger et al. (2008), estimated as the convex hull volume of the unique trait combinations in a community using the “dbFD” function of the package “FD” version 1.0-12.1 (Laliberté et al., 2014).

Divergence indices

We estimated the phylogenetic mean pairwise distance index (PD^{MPD}) and the phylogenetic mean nearest taxon distance index (PD^{MNTD}) (Webb et al., 2002). The PD^{MPD} measures the average phylogenetic distance between all species pairs in a community, reflecting deeper phylogenetic branching patterns. The PD^{MNTD} measures the average distance between a species and its closest relative in a community, reflecting evolutionary dynamics of closely related species at branch tips, which can be critical to untangling changing species interactions with other

species and their environments (Graham et al., 2018). We derived functional divergence indices (FD^{MPD} and FD^{MNTD}) analogously as the phylogenetic indices after converting the species traits to a trait distance matrix (Webb et al., 2002, 2008), thus capturing ecological processes specific to nearest-neighbor and community-wide relationships. We further estimated the incidence-based distance-based functional divergence index from Vileger et al. (2008) to quantify how species were divergent from the center of the community functional space/volume.

Community structure indices

The standardized effect sizes of the phylogenetic divergence indices (PD^{MPD} and PD^{MNTD}) and functional divergence indices (FD^{MPD} and FD^{MNTD}) were used to estimate phylogenetic community structure indices, including the phylogenetic and functional net relatedness index (PD^{NRI} and FD^{NRI} , respectively) and phylogenetic and functional nearest taxon index (PD^{NTI} and FD^{NTI} , respectively). These indices were aimed at assessing whether species assembling in communities differed from random assemblages (null models) based on their evolutionary (phylogenetic) relationships and ecological functions (Webb et al., 2002; Webb & Donoghue, 2004). The net relatedness indices represent the average relatedness between all possible pairs of taxa in a community, whereas the nearest taxon indices represent the mean nearest neighbor in a community (Webb et al., 2008). They were estimated by comparing PD^{MPD} , PD^{MNTD} , FD^{MPD} , and FD^{MNTD} to null-model communities generated using the independent swap algorithm (Gotelli & Entsminger, 2003) with 10,000 randomizations and 1000 iterations as implemented using the “ses.mpd” and “ses.mntd” functions in “picante” version 1.8.2 (Kembel et al., 2010). The resulting values were multiplied by -1 to obtain PD^{NRI} , PD^{NTI} , FD^{NRI} , and FD^{NTI} (Kembel et al., 2010; Webb et al., 2008): Negative values indicate that species in the community are less closely related than expected by chance (Webb et al., 2002, 2008), while positive values indicate that species are more closely related or more similar in their phylogenetic history or functional traits than expected by chance. For comparison, we also estimated the phylogenetic species evenness index of Helmus et al. (2007) and the incidence-based distance-based functional evenness index from Vileger et al. (2008).

Environmental covariates

We first explored the effect of spatial scale on the predictive influence of environmental and anthropogenic variables on species diversity and distribution patterns using several environmental, climatic, and anthropic variables (Ceballos &

Ehrlich, 2006; Davies et al., 2011; Hawkins et al., 2003; Jenkins et al., 2013) and then applied correlation tests to ensure that the covariates retained for tests were least correlated ($|<0.7|$ Pearson correlation coefficient) (Graham, 2003). Our pre-analysis confirmed that the coverage and heterogeneity of these variables were sufficiently representative of local conditions (see Appendix S1: Figures S7 and S8) (MEWNR, 2015; Philip et al., 2015), suggesting that they could reliably model environmental effects on species biogeography across the range of spatial scales of our study. The variables included the annual temperature average (TAM), annual temperature seasonality (TAS), annual average precipitation (PAM), seasonality in annual precipitation (PAS), terrain ruggedness index (TRI), actual annual evapotranspiration (AET), normalized difference vegetation index (NDVI), and human ecological footprint (HFP). The PAM, PAS, TAM, and TAS data were obtained from near-current bioclimatic variables (Fick & Hijmans, 2017; Karger et al., 2017). The AET was extracted from Running et al. (2021) and the NDVI from Didan (2015). The TRI was calculated from the digital elevation model layer (extracted from NASA's SRTMGL1 v003 1-arc second (about 30 meters) resolution DEM layer) using the "Terrain Ruggedness Index" raster analysis function in QGIS (QGIS Development Team, 2024). The HFP was extracted from the averaged Venter et al. (2018) and Kennedy et al. (2020) datasets for a more enhanced representation of human activity proliferation in wildlife areas. The PAM, PAS, TAM, TAS, and AET were used as climatic variables to test the water-energy availability hypotheses (Brown, 2014; Hawkins et al., 2003). The NDVI represents habitat primary productivity, another essential driver of mammalian diversity distribution patterns in the tropics. The TRI plays a crucial role in determining mammal habitat use and density, such as providing cover for prey and predators and influencing species form, function, and behavior in response to abiotic and biotic factors and disturbances (Riley et al., 1999; Whittaker et al., 1973). All variables were transformed using natural logarithm to normalize distributions. These global climatic datasets offer a relatively reliable depiction of the ecological conditions that affect mammal biodiversity patterns on a broad scale, as well as local microclimatic variations (Coelho et al., 2023). At a 30-arc second resolution (approximately 1 km² around the equator), they capture significant local geographic details, which is particularly valuable in regions with sparse data (Fick & Hijmans, 2017; Karger et al., 2021).

Statistical analysis

Associations between diversity indices and covariates across spatial scales were modeled using generalized

mixed linear regression models. Because of the nested sampling design, we implemented spatially constrained models based on communities' centroid geographic coordinates (Dormann, 2007; Legendre, 1993). The models, implemented in R using the "spaMM" package version 4.3.0 (Rousset & Ferdy, 2014), included the environmental variables as fixed effects, specified a Matérn correlation function and smoothness parameter (Matérn (1|Longitude + Latitude)), a Gaussian distribution, and identity link function due to normally distributed response and predictor variables, and used the restricted maximum likelihood (REML) to reduce bias in the estimation of variance components (Meyer, 1991). By integrating spatial coordinates into the correlation function, the models structurally factored in the spatial layout and the distances among the study units, providing a framework that captured the spatial dependencies in the data. We verified model assumptions, including the normality of residuals and random effects, the linearity of relationships, variance homogeneity, and the absence of multicollinearity, using the "performance" package version 0.10.4 (Lüdtke et al., 2021). The effects were summarized using Akaike information criterion (AIC) and R^2_a (adjusted coefficient of determination, R^2) (Burnham & Anderson, 1998; de la Sancha et al., 2023), with lower AIC and higher R^2_a values interpreted as strong diversity-covariate associations and better model performances (Burnham & Anderson, 2004). The parameters were processed and summarized using the "parameters" R package version 0.21.6 (Lüdtke et al., 2020).

Processing of literature-assembled nationwide data

We corroborated the field survey datasets with a nationwide sampling from the literature, compiling a species checklist matching the ordinal ranks recorded from the field surveys. We applied the nested quadrant sampling (SNQ) method to define spatially variable sampling units where a similar spatial extent and shape of the sampling window across scales is maintained, thus providing more accurate estimates of expected SR for a plot randomly located in a given area. In total, we established 516 sampling grids/plots spanning 16 spatial grains (independent datasets): 1, 10, 20, 40, 60, 80, 100, 200, 300, 400, 500, 600, 700, 800, 900, and 1000 km² (Appendix S1: Figure S9). These scales were defined by leveraging our previous fieldwork in Kenya and with reference to ecologically meaningful spatial scales at which small mammal community ecology processes are observable (Bowman et al., 2000; Storch et al., 2012). The 516 plots were the

maximum number of squared grids spanning the extent of Kenya at a 1000-km² grid size, with the subsequent 15 spatial scales established by first defining the centroids of the 1000-km² grids and then creating buffers of spatially varied radii lengths (back-calculated to correspond to grid sizes using the formula of pi multiplied by the squared radius [$A = \pi r^2$]).

For species composition, we used the annotated species checklist of mammals of Kenya by Musila, Monadjem, et al. (2019) as the backbone, from which we retained 147 small mammal species in the orders Rodentia, Eulipotyphla, and Macroscelidea, with body mass <10 kg (Onditi et al., 2025: data tab. 3). Species ranges and occurrences were obtained from the International Union for Conservation of Nature (IUCN, 2024) and the Global Biodiversity Information Facility (GBIF.org, 2023). The distributions of species without range or occurrence data in these databases were resolved by creating new range polygons based on the literature accounts of their distribution. The molecular phylogeny of the species retained in the community dataset was extracted from Kumar et al. (2017), with species missing from the extracted phylogeny randomly assigned to the subtree descended from the most recent common ancestor of the genus members using the “add.species.to.genus” function in the R package “phytools” version 1.2.0 (Revell, 2011, 2024). Because species ecologies and evolutionary histories are significantly impacted by geographical characteristics (Wiens, 1989), we classified the grids into four geographically weighted clusters (coastal, northeast, northwest, and southwest) based on species phylogenetic affinities using the “phyloregion” R package version 1.0.6 (Daru, Karunarathne, et al., 2020). Subsequent analyses were performed separately for each geographical cluster and an unpartitioned “combined” dataset.

The pipeline for extracting functional traits and environmental predictors, deriving diversity indices, and implementing statistical analysis were similar to those used for the new field survey dataset.

RESULTS

The 139 “species” identified in the new field survey dataset were distributed across three orders—Rodentia (78.4%), Eulipotyphla (20.9%), and Macroscelidea (0.7%), five families—Gliridae, Macroscelididae, Muridae, Nesomyidae, and Soricidae—and 28 genera (Onditi et al., 2025: data tab. 2). On the other hand, the nationwide literature checklist included 147 species in 52 genera, 14 families, and three orders—Rodentia (70.75%), Eulipotyphla (25.85%), and Macroscelidea (3.4%) (Onditi et al., 2025: data tab. 3).

Spatial scale effect on diversity distribution patterns

Associations between diversity indices varied across the sampled grains, with related facets (i.e., richness, divergence, and community structure) exhibiting similar topologies (Appendix S1: Figures S4–S6). In the new field survey dataset, SR and divergence either monotonically increased or decreased as the spatial scale increased; the community structure indices exhibited nonmonotonic patterns (Figure 2; Appendix S1: Figure S2).

As the spatial scale increased, there was an increase in community-wide phylogenetic divergence (mean pairwise distances between taxa: PD^{MPD} and FD^{MPD} increased) but a decrease in the divergence between nearest congeners (tree tip relationships: PD^{MNTD} and FD^{MNTD} decreased; Figure 2). Sites increasingly comprised clustered communities as the spatial scale increased (Figure 2). Distinct differences emerged between regions, with the largest scale (sites) clustered for both indices except in Mt. Kenya and Shimba-Gongoni for the PD^{NRI} and Loita for the PD^{NTI} , whereas the smaller scales always deviated less from random expectations.

The functional community structure indices (FD^{NTI} and FD^{NRI}) were clustered across spatial scales except in Mt. Kenya, Mau, and Loita, albeit with no apparent systematic variability across the spatial scales. Overall, the tendency of species to be overdispersed generally decreased with decreasing spatial grain size, that is, from sites, localities, transects, to traplines, with notable deviations between indices (Figure 2). For example, in Mt. Kenya, the basal-weighted indices (PD^{NRI} and FD^{NRI}) showed congruent distributions where the large scales were more overdispersed (extreme negative values), and smaller scales were less overdispersed (values closer to zero). On the other hand, PD^{NTI} patterns suggested increasingly clustered assemblages in larger areas, whereas for FD^{NTI} , overdispersion decreased with increasing scale (Figure 2). In the nationwide literature-assembled dataset, species, phylogenetic, and functional richness generally increased from small to larger scales, and divergence decreased as the spatial scale increased (Figure 3; Appendix S1: Figure S3). These patterns were more clearly reflected by the community structure indices, further demonstrating that communities were increasingly clustered at larger spatial scales and overdispersed at local small scales.

Some notable tip-weighted versus basal-weighted differences between regions emerged. For instance, the PD^{NRI} decreased with scale in the *coast* region but increased in the east and west regions and was relatively flat in the northwest region, contrasting with the PD^{NTI} , which had hump-shaped and inverted hump-shaped curves in the coast and west regions, respectively, but

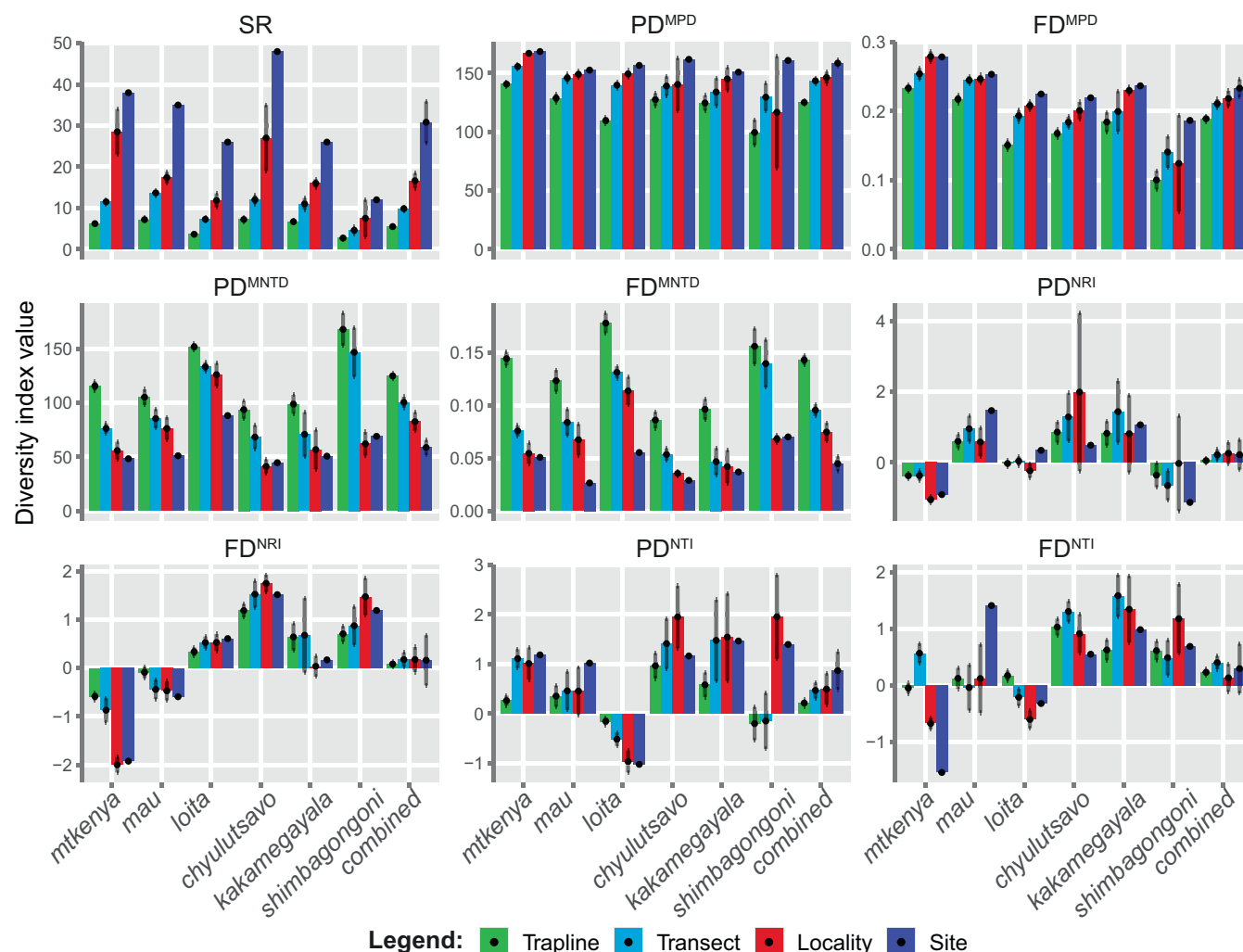


FIGURE 2 Distribution of diversity indices at various spatial scales across the sampled sites. The bars and points indicate the mean values, with 95% CIs around the mean. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations and Appendix S1: Figures S4–S6 for the principal components and correlation association analyses.

increased in the east and west regions. On the other hand, the FD^{NRI} generally decreased with increasing spatial scale in the coastal and southwestern regions but increased in the northeast and northwest regions, nearly mirroring the FD^{NTI} patterns, except in the northwest, where the FD^{NTI} presented a somewhat inverted hump-shaped curve as spatial scale changed (Figure 3; Appendix S1: Figure S3).

Spatial scale influence on diversity–environment association patterns

Associations between diversity indices and environmental predictors increased as the spatial scale increased, with R^2_a consistently being highest at larger scales (Figure 4a; Onditi et al., 2025: data tab. 4), and AIC having the opposite relationship (Figure 4b; Onditi

et al., 2025: data tab. 5). Similarly, these associations increased or remained stable as spatial scale increased in the nationwide dataset, albeit with different topologies between regions and indices (Figures 5 and 6). Across regions, SR and functional diversity indices' R^2 values generally increased as spatial scale increased (Figure 5). In the coastal region, associations were relatively stable and increasingly stronger with increasing scale, whereas in the inland regions (northeast, northwest, and southwest), there was pronounced variability and stronger scale dependencies in the diversity metrics (Figures 5 and 6). The relative influence of individual predictors was highly irregular across spatial scales, and correlations were generally stronger at larger scales and weaker at smaller scales, with the strength of these associations and model performances across spatial scales notably more systematic between regions than in diversity indices (Onditi et al., 2025: data tabs. 4 and 5).

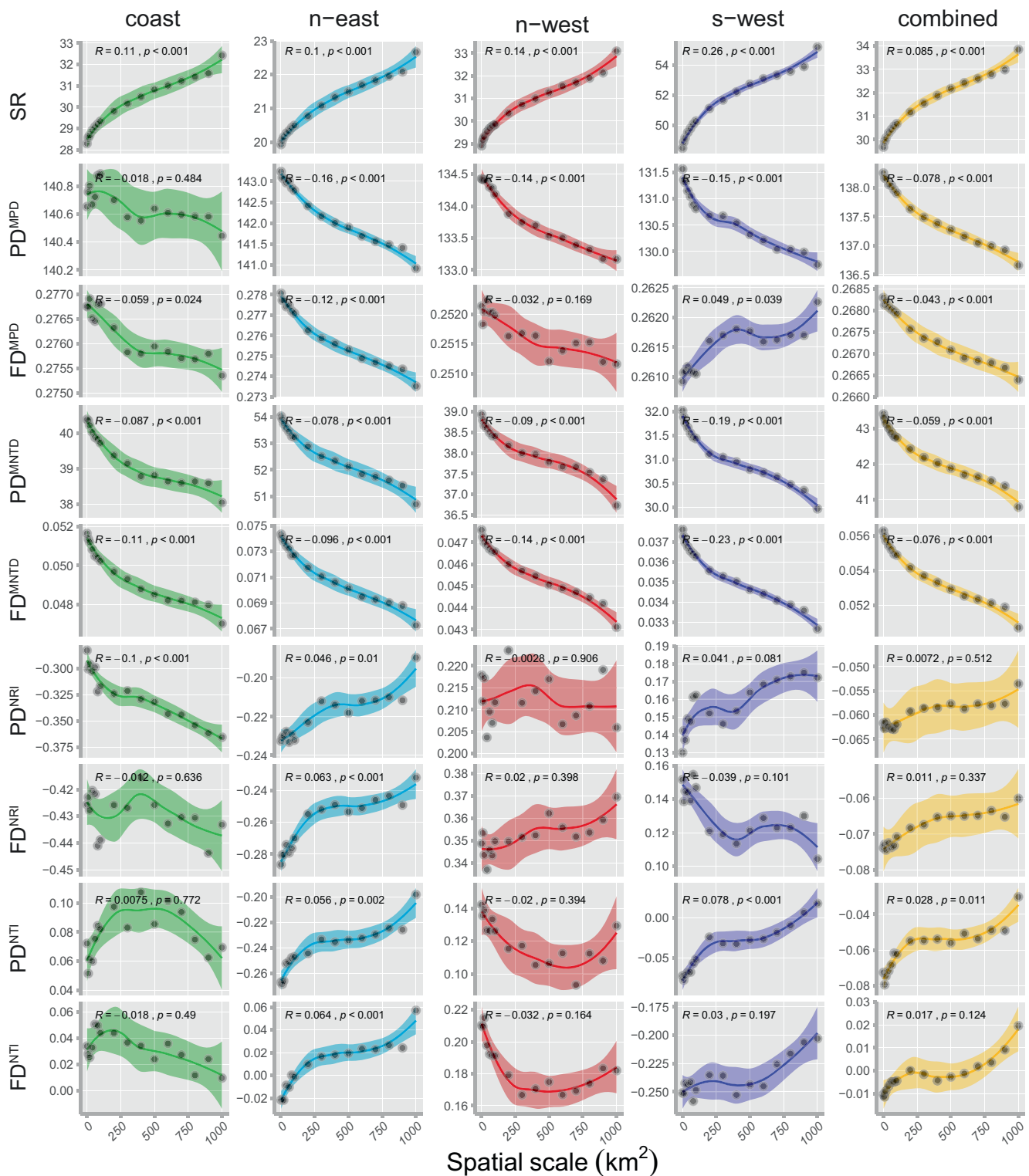


FIGURE 3 Distribution of diversity indices at variable spatial scales using a nationwide literature-assembled dataset. The points represent the mean values, the smooth curves are derived from local polynomial regression, and shaded areas around them represent the 95% CI from the SE of the estimated response. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations, Appendix S1: Figures S7 and S8 for the principal components analysis and correlational association tests, and Appendix S1: Figure S9 for the geographical illustrations of regions.

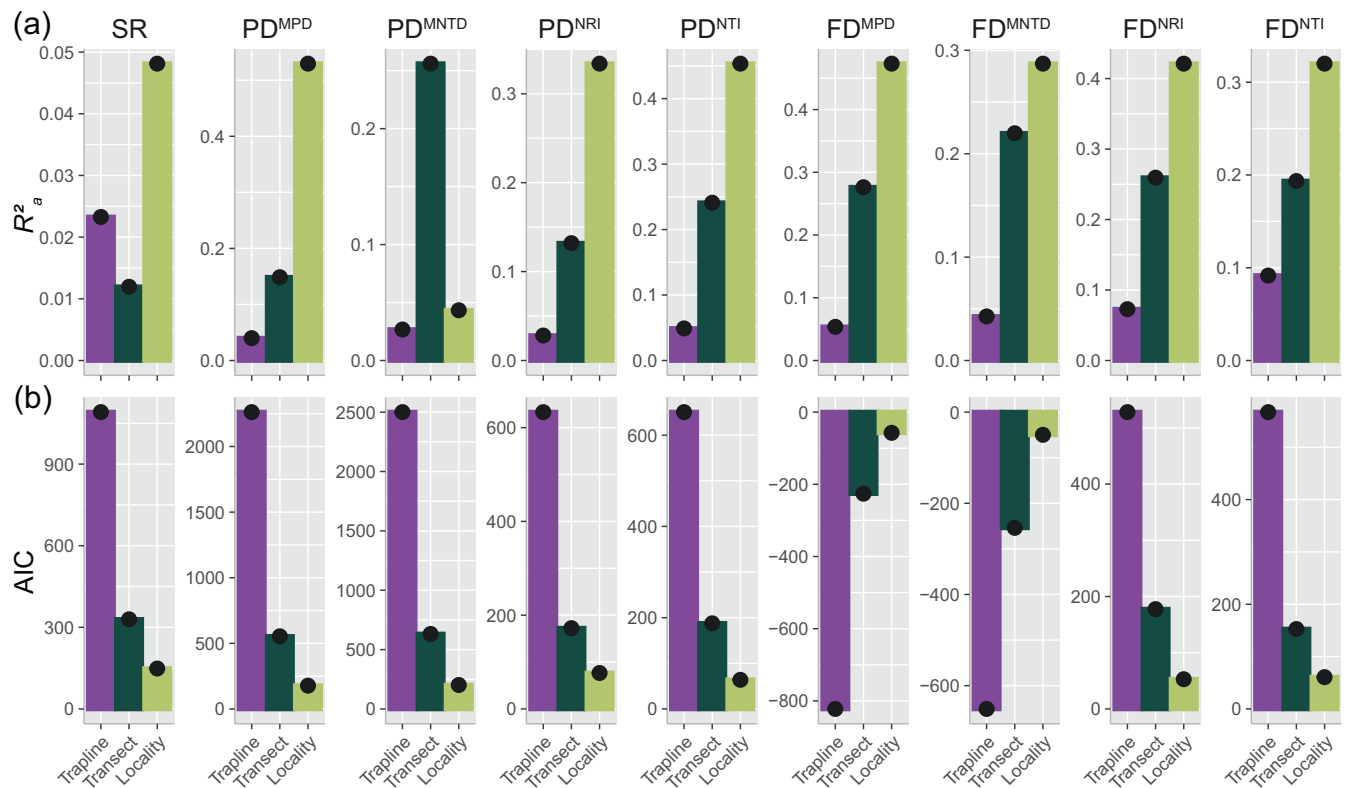


FIGURE 4 Influence of sampling spatial scale on the strength of diversity–environment associations across field survey sites and diversity indices. The bar and point-line plots depict how association strength changes with spatial scale, based on two metrics: (a) the proportion of variance in diversity explained by the model (adjusted R^2 , R^2_a) and (b) Akaike information criterion (AIC). Higher R^2_a values indicate better model performance, and lower AIC values indicate better model fit. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations.

Spatial scale influence on diversity–human ecological footprint association patterns

The distribution of human ecological footprint (HFP) differed as the spatial scale changed and exhibited distinct differences across the studied sites, with variable rate of change and range of HFP values (Figure 7a,b). While HFP intensity increased with spatial scale in the nationwide dataset (Figure 7b), differences emerged in the new field survey data. For instance, in the combined dataset, HFP was higher when estimated at larger scales (site and locality) than at smaller scales (trapline and transect), contrasting with observations at individual site levels, such as in the Mau Forest where HFP was greater at smaller scales than at larger scales (Figure 7a). The spatial scale influence on diversity–HFP associations was stronger at larger scales and weaker at smaller scales in the new field survey dataset (Figure 7c,d). However, in the nationwide dataset, these associations were distinct and complex between regions and indices (Figures 8 and 9), although generalizable to be stronger at larger sampling grains based on the frequency of increasing explained variances and model performance across spatial scales (Figures 8 and 9).

DISCUSSION

We explored the relationships between spatial scale and various taxonomic, phylogenetic, and functional diversity indices—richness, divergence, and community structure—to understand how they affect observed small mammal diversity and distribution patterns. Our study revealed systematic variations in how sampling grain differences influence observed diversity patterns and the impact of environmental factors on these patterns, both in primary field survey data and in literature-compiled data.

Spatial scale inherency in diversity distribution patterns

We observed a consistent increase in taxonomic, phylogenetic, and functional richness as spatial scale increased, a pattern attributable to the nested nature of smaller scales within larger scales. Generally, the relationships between SR and richness-dependent indices and spatial scale variations align with the well-established species–area relationship hypotheses; that is, larger natural areas harbor

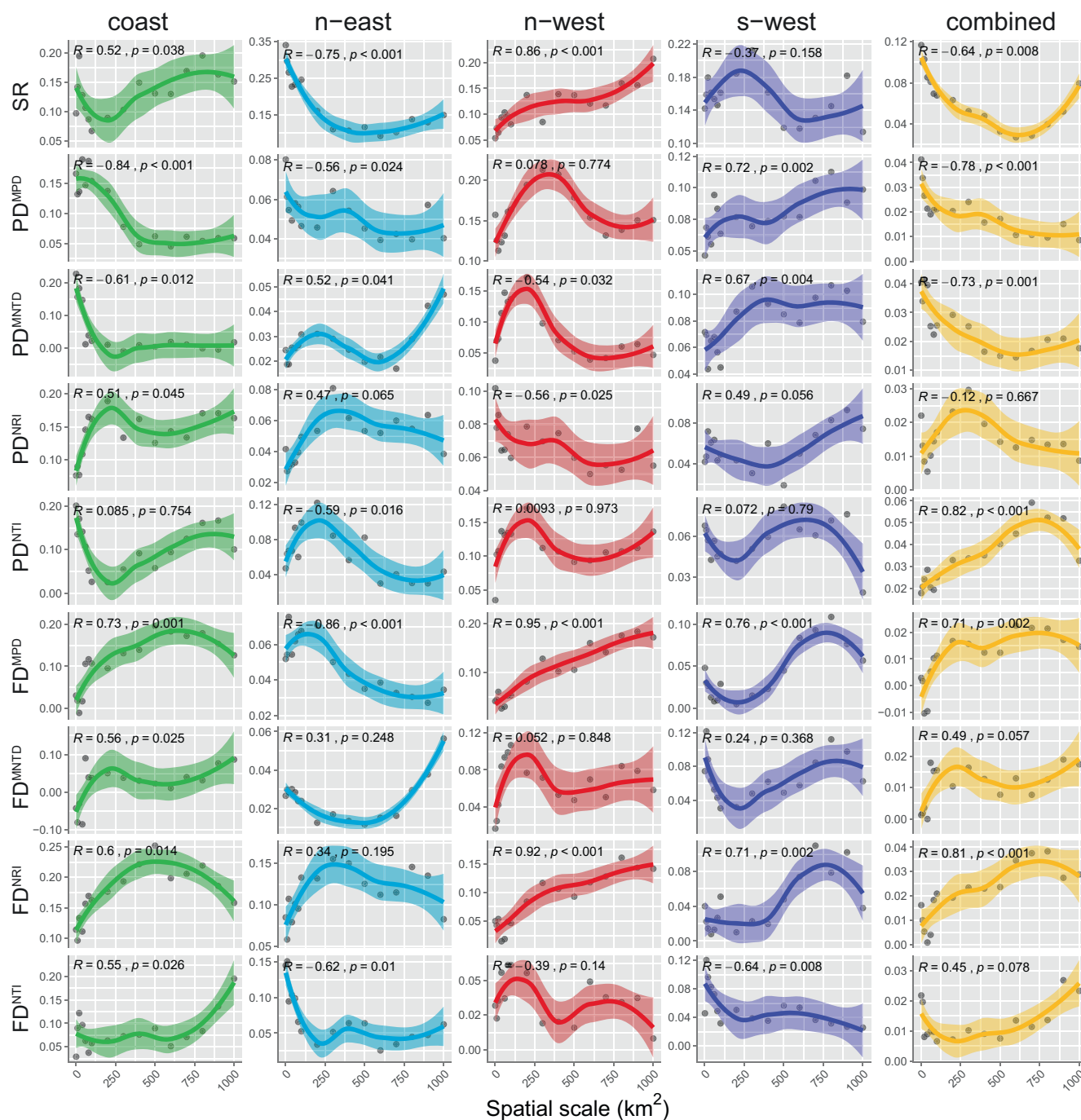


FIGURE 5 Influence of sampling spatial scale on diversity–environment associations across regions in the nationwide dataset, illustrated by the proportion of variance in diversity explained by models (adjusted R^2 , R^2_a). Smooth curves are fitted with local polynomial regression, and the shaded areas indicate the CIs derived from the SE of the estimated response. The correlation coefficients and corresponding p values, based on Pearson's correlation, are shown for each relationship. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations and [Appendix S1: Figure S9](#) for the geographical illustrations of regions.

more species than smaller areas do (Hutchinson, 1953; Levin, 1992; Rosenzweig, 1995, 2005; Whittaker et al., 2001; Willis Katherine & Whittaker Robert, 2002). On the other hand, the patterns depicted by the divergence and community structure indices highlighted the complex nature of species assembly when accounting for

their ecological functional and evolutionary histories. More dispersed assemblages at smaller spatial scales support the theoretical hypothesis that at local scales, strong interspecific competition causes repulsion between closely related species, leading to phylogenetic and trait overdispersion (Scherson & Faith, 2018; Webb

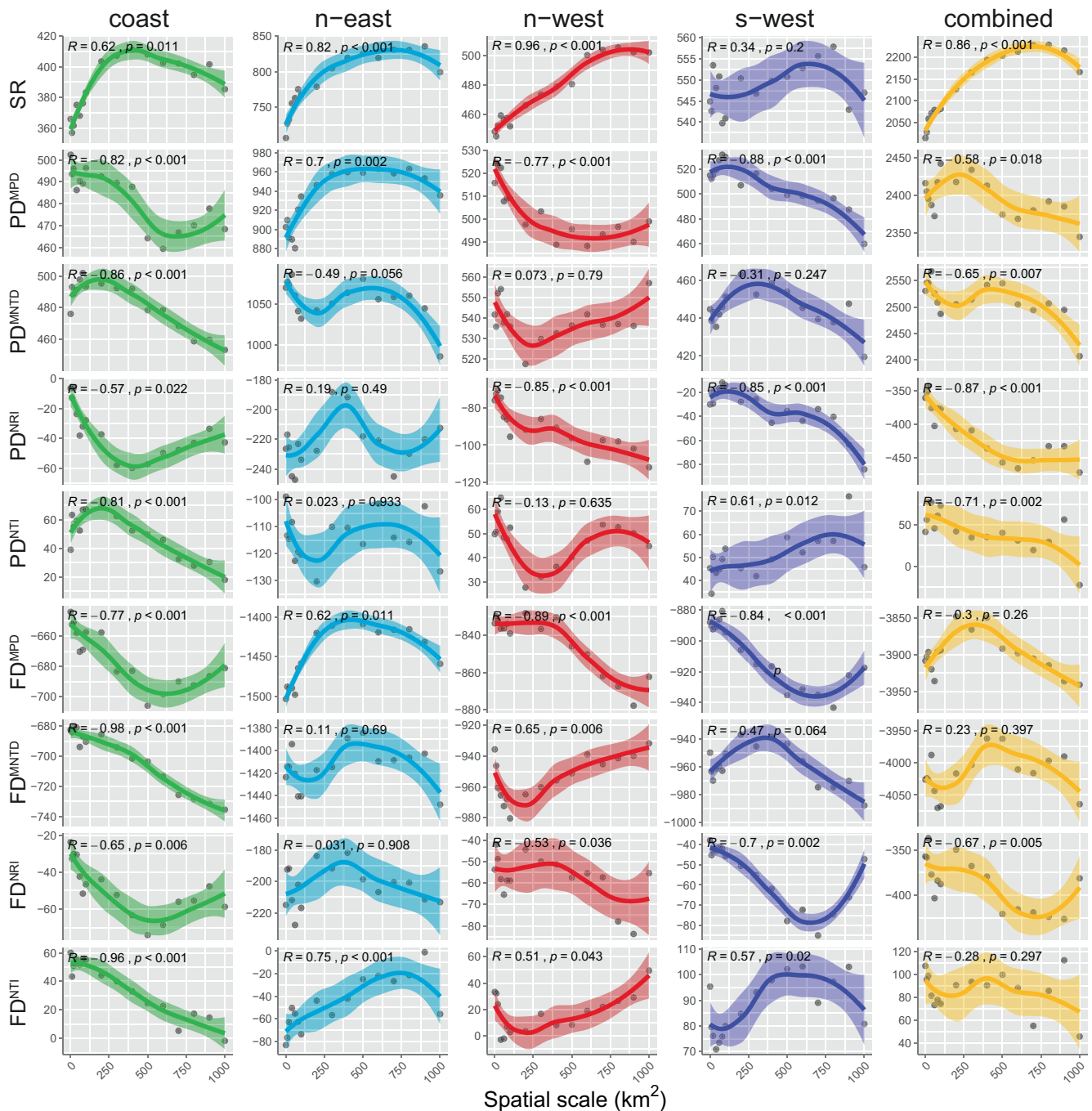


FIGURE 6 Influence of spatial sampling scale on diversity–environment associations across regions in the nationwide dataset, assessed using estimated prediction error to evaluate model performance (Akaike information criterion [AIC]). Smooth curves represent local polynomial regression fits, with shaded areas indicating 95% CIs derived from the SE of the estimated response. Pearson's correlation coefficients and corresponding p values are provided for each relationship to quantify the strength and significance of the associations. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations and [Appendix S1: Figure S9](#) for the geographical illustrations of regions.

et al., 2002). For small mammals such as shrews and rodents, competitive interactions are more pronounced at smaller spatial scales and weaker at larger scales. Moreover, the distinct deviations in the tempo by which species form and are maintained in an area seem to conform to nonrandom variabilities in ecological and

evolutionary opportunities that are influenced by the spatial size of the community in rather complex ways based on environmental context (Graham et al., 2014, 2018; Pyron et al., 2015). Our findings underscore the critical role of spatial scale in understanding species distributions and assembling along environmental gradients (Truchy

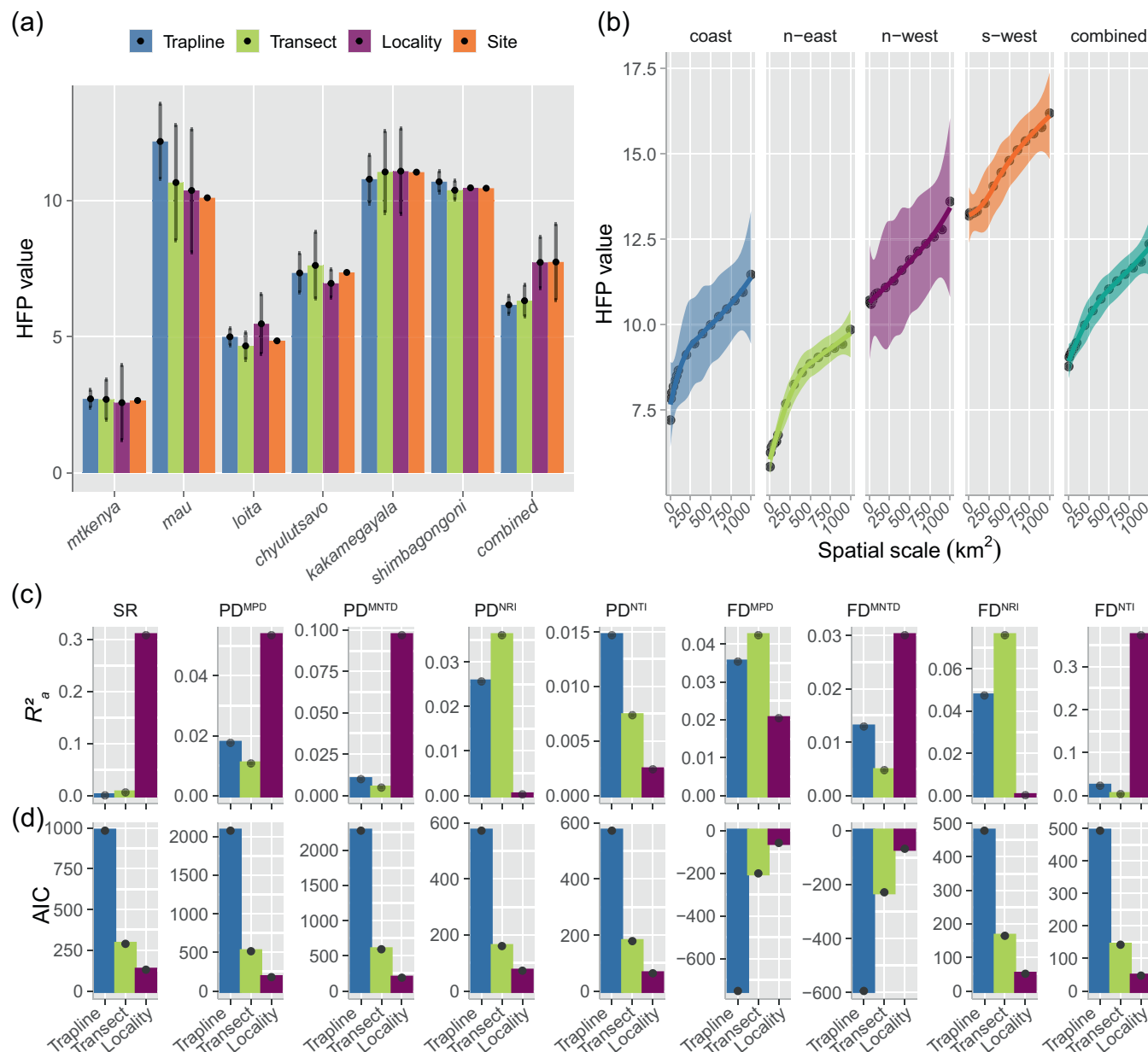


FIGURE 7 The effect of sampling spatial scale on the association between the human ecological footprint (HFP) and small mammal diversity patterns: (a) variability in HFP estimates averaged across survey sites at different spatial scales; (b) variability in HFP estimates averaged across nationwide regions, with smooth curves fitted using local polynomial regression and shaded areas representing 95% CIs derived from the SE of the estimated response—the geographical configuration of regions is shown in Appendix S1: Figure S9; (c) proportion of variance in diversity patterns explained by models (adjusted R^2 , R^2_a) as a function of sampling spatial scale, indicating the impact of scale on HFP-diversity associations, and (d) variations in model performance, assessed using prediction error (Akaike information criterion [AIC]), highlighting the influence of spatial scale on the strength and accuracy of HFP-diversity relationships. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations.

et al., 2019). Conducting comprehensive field surveys with broader spatial sampling grains is the best approach to accurately capture biodiversity patterns and their driving factors, particularly when studying evolutionarily distinct and ecologically significant species that are likely to have heightened sensitivity to changes in spatial scale. Such insights are vital for making informed conservation decisions, helping determine the spatial focus of diversity

conservation efforts and the appropriate scales for these efforts. Particularly notable are the distinct shifts in community structure indices observed at the 250- to 500-km² scale in the nationwide dataset, which suggests that strategic decisions made at these scales may effectively balance species and ecosystem conservation with practical cost considerations (Fahrig et al., 2022; Henle et al., 2014; Onditi et al., 2023).

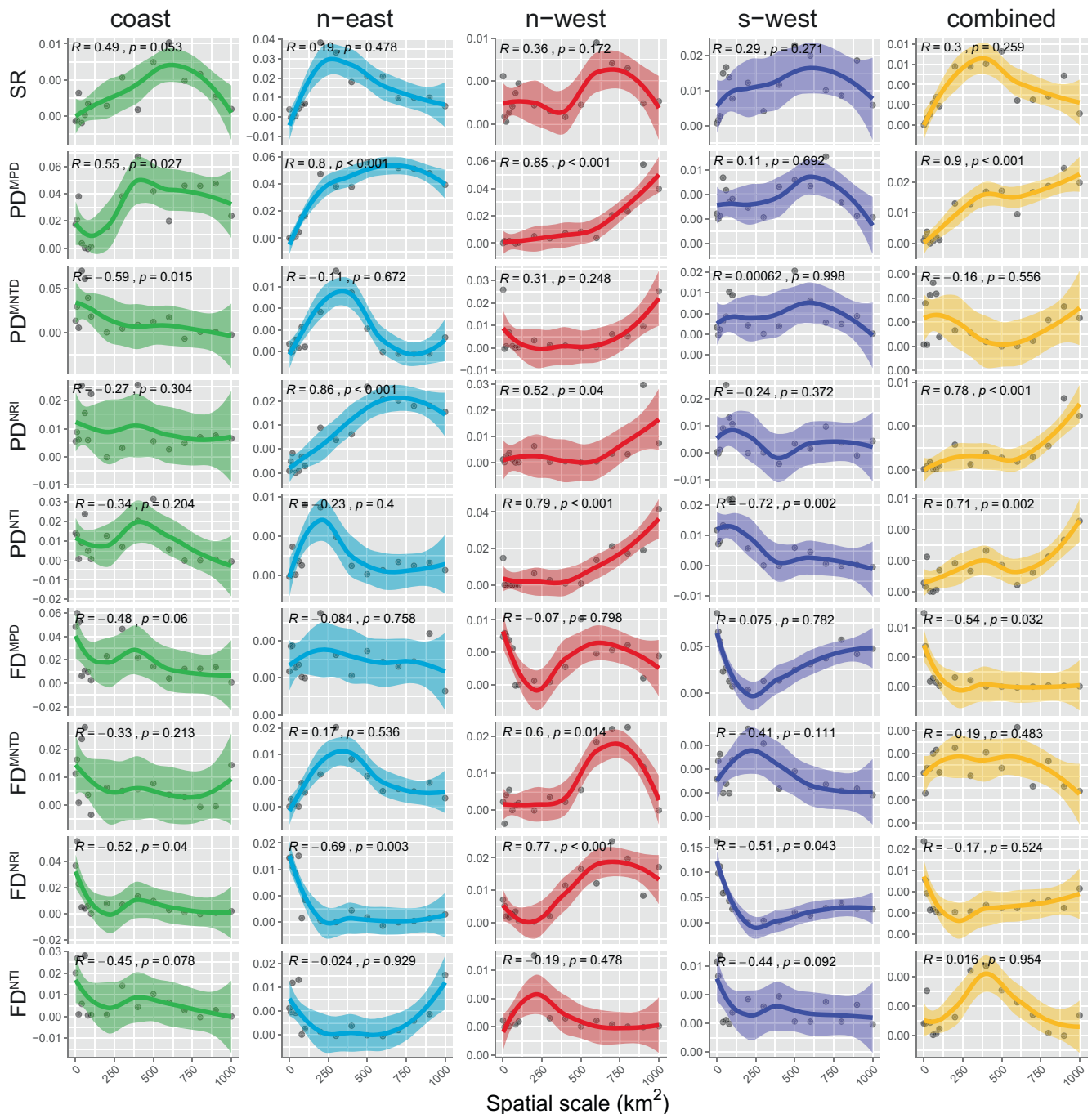


FIGURE 8 The impact of spatial scales on human ecological footprint (HFP) influence on small mammal diversity patterns in the nationwide dataset. The variability in proportion of variance in diversity patterns explained by HFP (adjusted R^2 , R_a^2) across regions when estimated at different spatial scales are shown. Smooth curves are derived from local polynomial regression, and shaded areas around them represent the 95% CI derived from the SE of the estimated response. Pearson's correlation coefficients and corresponding p values are provided for each relationship to quantify the strength and significance of the associations. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations and Appendix S1: Figure S9 for the geographical illustrations of regions.

Spatial inherency in geographic regionalization of species distribution patterns

We found that assemblages at larger scales were composed of taxa that are ecologically and evolutionarily

more related than would be expected by chance, results aligning with previous research (Buckley et al., 2010; Cardillo, 2011; Kamilar et al., 2015) and supporting the hypothesis that environmental filtering effects are stronger at these scales. Species poorly adapted to prevailing environmental conditions are systematically filtered out

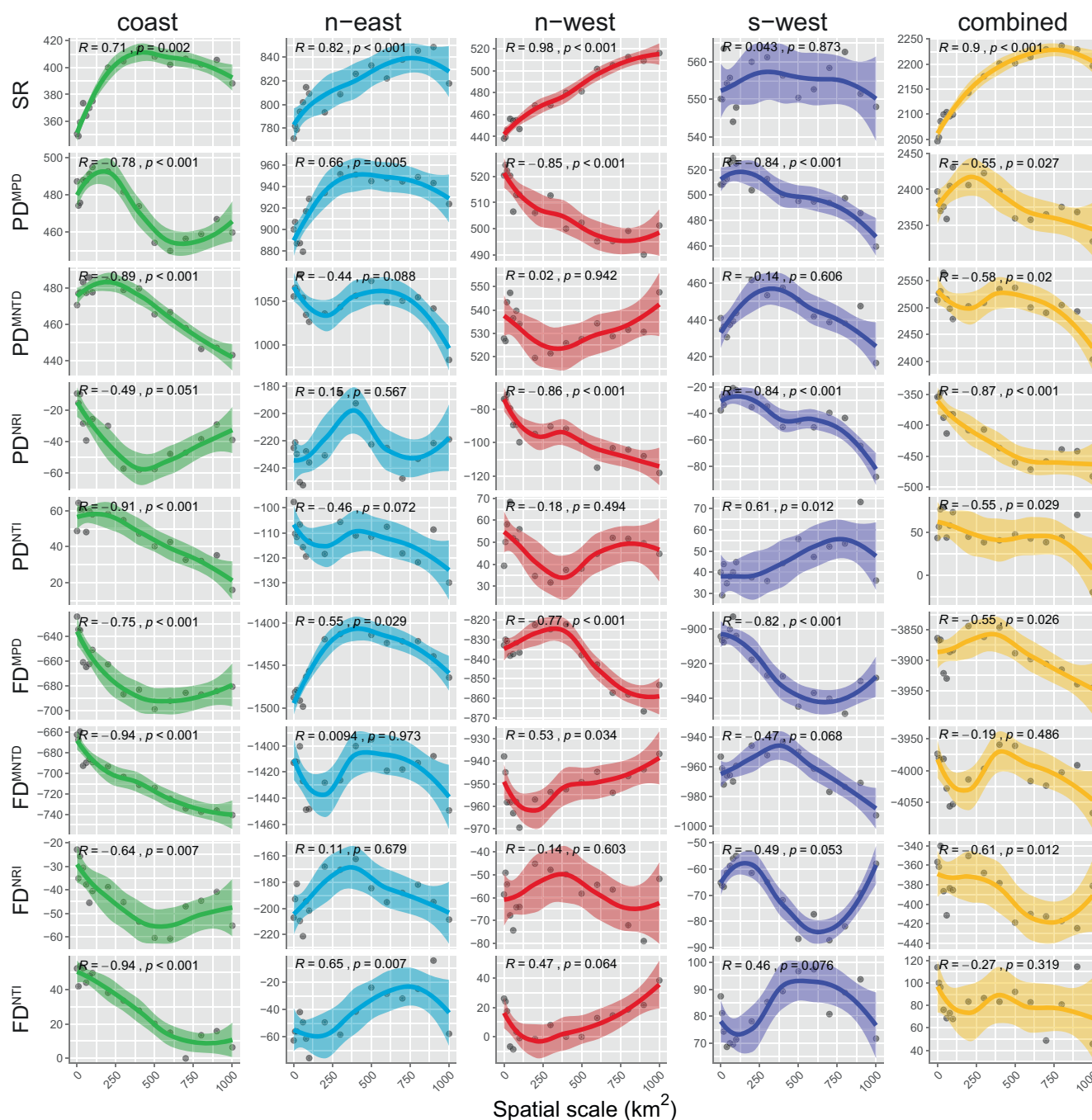


FIGURE 9 Influence of spatial scales on the effect of the human ecological footprint (HFP) on small mammal diversity patterns in the nationwide dataset. The variability in diversity–HFP model performance at different spatial scales, evaluated using prediction error (Akaike information criterion [AIC]), is shown across regions. Smooth curves represent local polynomial regression fits, with shaded areas indicating 95% CIs derived from the SE of the estimated response. Pearson's correlation coefficients and corresponding p values are included to quantify the strength and statistical significance of the associations. See [Methods: Diversity indices estimation](#) for explanations of diversity abbreviations and Appendix S1: Figure S9 for a geographical representation of the regions.

from communities, with species more likely to retain traits in their ancestral forms for lack of ecological or competition pressures to adapt them otherwise. According to our field survey dataset, each study site likely serves as a pool for species uniquely adapted to

local conditions. The systematic regionalization patterns in diversity distributions and diversity–environment associations at different spatial scales underscore the importance of careful consideration when generalizing pattern drivers across broader species pools. The nonrandom

assembly of species across different sites and regions suggests that evolutionary and ecological opportunities for species are constrained differently within specific biogeographical boundaries. A clear contrast in functional and phylogenetic clustering emerged between the study sites. For example, the overall functional and phylogenetic clustering (increasingly positive values) in the Chyulu-Tsavo, Kakamega-Yala, and Shimba-Gongoni communities contrasted with the tendency of the Mt. Kenya, Mau, and Loita communities to overdispersion (increasingly negative values) or randomness (values closer to zero) for both tip-weighted and basal-weighted relationships. In both cases, the patterns appear to be related to the species compositions. The MOTUs from Mt. Kenya comprised 15 genera, Mau 17 genera, and Loita 18 genera, while 14 genera were recorded in Chyulu-Tsavo, 12 in Kakamega-Yala, and 9 in Shimba-Gongoni. The community structure patterns can therefore be explained by the clustered sites being inhabited by few “specialist” taxa uniquely habituated to the ecosystems owing to overall stronger environmental filtering. This argument for strong environmental filtering in Chyulu-Tsavo and Shimba-Gongoni, for example, is further explicable by the fact that these ecosystems are characterized by low habitat productivity (low forest cover and density, low annual precipitation, high annual temperature, higher seasonality in precipitation, and temperature than Mau Forest and Mount Kenya) (see Appendix S1: Figures S7 and S8). Unique observations such as the community structure indices in the coastal region, which, albeit in contrast with the general expectation of clustering at large scales and overdispersion at smaller scales (Scherson & Faith, 2018; Webb et al., 2002), seem ecologically and evolutionarily connected. When sampling a larger area does not necessarily increase phylogenetic distances between species, more species in these areas are likely to have close basal relationships due to niche conservatism, as is typical in ecosystems where environmental conditions impose stronger filtering influences on species assembly (Webb et al., 2002). The distinct climate of coastal Kenya fosters highly localized faunal and floral diversity and endemism due to convergent adaptive responses to unique environmental conditions, resulting in communities of more closely related species than are expected by chance (Cardillo, 2011; Olalla-Tárraga et al., 2017). Overall, our findings highlight that the influence of spatial scale on various taxonomic, phylogenetic, and functional diversity patterns systematically varies between regions, thus increasing our understanding of species assembly dynamics and how biodiversity patterns across large geographical areas are mirrored in more localized environments.

Variation in diversity–environment associations across spatial scales

The strength of associations between diversity indices and environmental variables systematically varied between spatial scales, with models performing better and predicting stronger diversity–covariate associations at larger scales than at smaller spatial scales. These associations are underlain by species diversity and distribution at larger spatial scales being more strongly influenced by greater heterogeneity in environmental conditions such as climate, topography, and habitat productivity (Ackerly, 2003; Bae et al., 2018; Cavender-Bares et al., 2006; Daru, Farooq, et al., 2020; Field et al., 2008; McBride et al., 2014; Pyron et al., 2015). Larger spatial scales typically encompass greater environmental heterogeneity, offering more exploitable niches and capturing broader ecological processes and interactions, which collectively exert a stronger influence on diversity patterns. However, regional variations suggest that local environmental and ecological conditions can significantly modulate these associations, necessitating region-specific approaches in biodiversity studies and conservation planning and underscoring the importance of considering spatial scales in ecological research to assess and predict biodiversity patterns with enhanced accuracy.

Nonetheless, studies increasingly highlight contrasting observations have emerged, such as overdispersed or unchanging assemblages at larger scales and clustered assemblages at smaller scales. For instance, Delcros et al. (2014) did not find strong associations between species nestedness and environmental variables, suggesting strong biotic filtering at smaller microhabitat scales or strong abiotic filtering at larger scales among rodents and shrews in South African savannah biomes, leading them to argue in favor of stronger habitat filtering at the local scale. Similarly, Qian and Kissling (2010) found that climate and habitat heterogeneity only weakly affected vertebrate SR at broader regional than local scales among terrestrial vertebrate and vascular plants in China. Overall, few studies have focused on the influence of spatial scale on environmental correlations with phylogenetic and functional community structure. González-Maya et al. (2016) reported that climate and productivity influence functional diversity more strongly at regional scales than at smaller scales for SR, which our results corroborate. We found that the correlation between the environmental variables and phylogenetic and functional diversity indices generally increased as the scale increased but was highly variable across regions. These variations, however, were geographically systematic, suggesting that strong synergism between topographic heterogeneity and climate across spatial scales is a dominant driver of species diversity and distribution (e.g., Rahbek &

Graves, 2001). The rugged topography of central and western Kenya implies that varying the sampling window can result in a significant change in the range of ecosystems and a corresponding change in species composition and assembly dynamics. On the other hand, the relatively flatter northeastern and coastal regions of Kenya imply that only large changes in the spatial scale of sampling extent may result in notable heterogeneity in ecosystems and species composition.

In the dedicated analyses on how sampling grain variations impact associations between HFP and diversity patterns, we recorded complex interactions between HFP, diversity indices, and spatial scale variations, similarly underscoring the need for region-specific research and conservation strategies that account for geographical differences and the appropriate spatial and methodological scales to address the challenges posed by human-induced environmental changes effectively. While the heterogeneity in HFP distribution across sampling grain sizes and regions may be attributed to differing levels of development, conservation policies, and human activity intensities across Kenya, our results reiterate the importance of considering scale when estimating human activity impacts on biodiversity. The spatial scales corresponding to the strongest associations could be pivotal in designing more effective species and ecosystem monitoring and conservation strategies, where a focus that is too broad or too narrow might miss significant patterns. The varying responses of different diversity indices to HFP also stress the complexity of ecological studies and the need for careful selection and interpretation of these metrics. Understanding which indices are more sensitive to human impacts can further guide more ecologically targeted research and conservation efforts.

Study caveats

Incorporating extensive primary field survey data alongside expert-modeled datasets can effectively address challenges related to site accessibility, sampling extent, and methodological variations (Levin, 1992; Wiens, 1989). This approach allows for a more nuanced, multi-scale perspective that refines analyses across diverse sampling grains. Nonetheless, it is critical to recognize that variations in how spatial scales are defined and interpreted across studies, geographic contexts, and species groups can yield inconsistencies and complicate comparisons. Moreover, inferences or projections based on our findings should be constrained with comparability in data types, resolutions, geographic conditions, and time windows. Future investigations could conduct

more targeted empirical tests to identify the most suitable scales for examining, interpreting, and applying spatial dimensions across species groups and geographical contexts, thereby advancing our understanding of dynamic ecological and evolutionary processes with enhanced accuracy.

CONCLUSION

We extensively investigated spatial scale influences on the biogeography of small mammals in Kenya using multiple diversity indices from new field survey data complemented with literature-assembled data across spatial scales. We found that across regions, although SR invariably increased as the spatial scale increased, there was a general increase in the tendency of species to be functionally or phylogenetically less related than expected by chance (clustered assemblages) as the spatial scale increased. Contrasting patterns among the sampled sites, regions, and indices suggest that the influence of spatial scale on various facets of diversity is somewhat random when geographical regionalization in species composition is not considered, complicating inferential generalizations. Although there were significant nonrandom effects of spatial scale on diversity distribution, suggesting an ingrained relevance to species assembly based on evolutionary and ecosystem functions, we did not record a consistent plateau of diversity distribution or diversity–environment associations across the sampled grain variations. The precise spatial scales at which environmental or competitive filtering gains prevalence as the main drivers remain unclear; pinpointing a specific spatial grain at which to conduct field studies should be informed by several factors unique to the species being studied, the geographical configuration of the study area, resources, and the target diversity facet (i.e., whether SR or phylogenetic/functional diversity aspects). The environmental variables were increasingly stronger filters of species diversity patterns as the spatial scale increased, corresponding to increased environmental filtering and decreased competitive interactions. These observations further support the famed species pool and environmental filtering hypotheses that the size of the species pool strongly influences the SR of the local community. At larger spatial scales, species persistence is more strongly determined by environmental heterogeneity, including climate, topography, and human influences, leading to communities comprising species that are more functionally and phylogenetically similar than would be expected by chance. Our study refines the understanding of species' ecological and evolutionary patterns and processes across spatial scales and could enhance the effectiveness of biodiversity management systems.

AUTHOR CONTRIBUTIONS

Kenneth Otieno Onditi conceptualized the idea, wrote the original draft, and reviewed and edited the manuscript. Wenyu Song contributed to conceptualization, data validation, and manuscript review. Zhongzheng Chen, Quan Li, Noé U. de la Sancha, and Simon Musila participated in the investigation and reviewed the manuscript. Xueyou Li and Xuelong Jiang validated the work, provided resources, supervised the project, and reviewed and edited the manuscript. Xuelong Jiang provided overall project supervision. All authors contributed to subsequent revisions and approved the final version for submission.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data and code (Onditi et al., 2025) are available from Figshare: <https://doi.org/10.6084/m9.figshare.25975360>.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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